

BANK DATABASE – SQL VIEWS

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Section: 7

STEP 1: Create Database

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: Bindhu Madhavi Local instance... Weeks2(TCL)* Joint and Operations* My Example* Praticse Example(Week2)*

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Schema

Administration

Information: No object selected

1 • CREATE DATABASE IF NOT EXISTS BankDB;

2 • USE BankDB;

Limit to 1000 rows

Output

Action Output

#	Time	Action	Message
1	18:15:33	CREATE DATABASE IF NOT EXISTS BankDB	1 row(s) affected, 1 warning(s): 1007 Ca
2	18:15:33	USE BankDB	0 row(s) affected

STEP 2: CREATE CUSTOMER TABLE

MySQL Workbench

Bindhu Madhavi Local instanc...

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Navigator: ds* Weeks2(TCL)* Joint and Operations* My Example* Praticse Example(Week2)*

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Per
- Administration

Information

No object selected

```
1 • CREATE DATABASE IF NOT EXISTS BankDB;
2 • USE BankDB;
3
4 • CREATE TABLE Customer (
5     Customer_ID INT PRIMARY KEY,
6     Customer_Name VARCHAR(100) NOT NULL,
7     Phone VARCHAR(15),
8     Email VARCHAR(100),
9     City VARCHAR(50)
10 );
```

Output

Action Output

#	Time	Action	Message
1	18:15:33	CREATE DATABASE IF NOT EXISTS BankDB	1 row(s) affected, 1 warning(s): 10
2	18:15:33	USE BankDB	0 row(s) affected
3	18:18:13	CREATE TABLE Customer (Customer_ID INT PRIMARY ...	Error Code: 1050. Table 'customer

STEP 3: CREATE ACCOUNT TABLE

MySQL Workbench

Bindhu Madhavi Local instanc... x

File Edit View Query Database Server Tools Scripting Help

Navigator: eregate Functions(Week2)* Week1* sql command.txt* Trigeers (Week2) SQL File 15*

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Per
- Administration

Information :

No object selected

```

1 • CREATE DATABASE IF NOT EXISTS BankDB;
2 • USE BankDB;
3 • CREATE TABLE Customer (
4     Customer_ID INT PRIMARY KEY,
5     Customer_Name VARCHAR(100) NOT NULL,
6     Phone VARCHAR(15),
7     Email VARCHAR(100),
8     City VARCHAR(50)
9 );
10
11 • CREATE TABLE Account (
12     Account_No INT PRIMARY KEY,
13     Customer_ID INT,
14     Account_Type VARCHAR(20),
15     Balance DECIMAL(12,2),
16     Branch VARCHAR(50),
17
18     FOREIGN KEY (Customer_ID)
19     REFERENCES Customer(Customer_ID)
20 );
21

```

Output :

Action Output

#	Time	Action	Message
19	18:25:58	CREATE DATABASE IF NOT EXISTS BankDB	1 row(s) affected, 1 warning(s): 1
20	18:25:58	USE BankDB	0 row(s) affected
21	18:26:13	CREATE TABLE Customer (Customer_ID INT PRIMAR...	Error Code: 1050. Table 'custom
22	18:26:43	CREATE TABLE Account (Account_No INT PRIMARY...	0 row(s) affected

Object Info S

STEP 4: CREATE BANK TRANSACTION TABLE

MySQL Workbench

Bindhu Madhavi Local instanc...

File Edit View Query Database Server Tools Scripting Help

Navigator: eragate Functions(Week2)* Week1* sql command.txt* Trigeers (Week2) SQL File 15*

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Per
- Administration

Information: No object selected

```

11 CREATE TABLE Account (
12     Account_No INT PRIMARY KEY,
13     Customer_ID INT,
14     Account_Type VARCHAR(20),
15     Balance DECIMAL(12,2),
16     Branch VARCHAR(50),
17
18     FOREIGN KEY (Customer_ID)
19     REFERENCES Customer(Customer_ID)
20 );
21 CREATE TABLE Bank_Transaction (
22     Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
23     Account_No INT,
24     Transaction_Type VARCHAR(20),
25     Amount DECIMAL(12,2),
26     Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
27
28     FOREIGN KEY (Account_No)
29     REFERENCES Account(Account_No)
30 );
31

```

Output

Action Output

#	Time	Action	Message
✓ 20	18:25:58	USE BankDB	0 row(s) affected
✗ 21	18:26:13	CREATE TABLE Customer (Customer_ID INT PRIMAR...	Error Code: 1050. Table 'customer' alrea
✓ 22	18:26:43	CREATE TABLE Account (Account_No INT PRIMARY...	0 row(s) affected
✓ 23	18:27:16	CREATE TABLE Bank_Transaction (Transaction_ID IN...	0 row(s) affected

STEP 5: CREATE LOAN TABLE

MySQL Workbench

Bindhu Madhavi Local instanc... x

File Edit View Query Database Server Tools Scripting Help

Navigator: eregate Functions(Week2)* Week1* sql command.txt* Trigeers (Week2) SQL File 15* SQLA

MANAGEMENT

- Server Status
- Client Connecti...
- Users and Privile...
- Status and Syste...
- Data Export
- Data Import/Res...

INSTANCE

- Startup / Shutdc...
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Per...
- Administration

Information : No object selected

```

21 • CREATE TABLE Bank_Transaction (
22     Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
23     Account_No INT,
24     Transaction_Type VARCHAR(20),
25     Amount DECIMAL(12,2),
26     Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
27
28     FOREIGN KEY (Account_No)
29     REFERENCES Account(Account_No)
30 );
31 • CREATE TABLE Loan (
32     Loan_ID INT PRIMARY KEY,
33     Customer_ID INT,
34     Loan_Type VARCHAR(30),
35     Loan_Amount DECIMAL(12,2),
36     Interest_Rate DECIMAL(5,2),
37
38     FOREIGN KEY (Customer_ID)
39     REFERENCES Customer(Customer_ID)
40 );
41

```

Output : Conte

Action Output

#	Time	Action	Message
21	18:26:13	CREATE TABLE Customer (Customer_ID INT PRIMAR...	Error Code: 1050. Table 'customer' already exists
22	18:26:43	CREATE TABLE Account (Account_No INT PRIMARY...	0 row(s) affected
23	18:27:16	CREATE TABLE Bank_Transaction (Transaction_ID IN...	0 row(s) affected
24	18:27:55	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	Error Code: 1050. Table 'loan' already exists

Object Info

STEP 6: INSERT CUSTOMER DATA

MySQL Workbench

Bindhu Madhavi Local instanc... x

File Edit View Query Database Server Tools Scripting Help

Navigator: eragate Functions(Week2)* Week1* sql command.txt* Trigeers (Week2) SQL File 15*

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Per
- Administration

Information : No object selected

```

39 REFERENCES Customer(Customer_ID)
40 );
41
42 INSERT INTO Customer
43 (Customer_ID, Customer_Name, Phone, Email, City)
44 VALUES
45 (101, 'BindhuMadhavi', '9876543210',
46 'bindhumadhavi@gmail.com', 'Hyderabad'),
47
48 (102, 'Priya Sharma', '9876543211',
49 'priya@gmail.com', 'Vijayawada'),
50
51 (103, 'Arjun Reddy', '9876543212',
52 'arjun@gmail.com', 'Bangalore'),
53
54 (104, 'Sneha Rao', '9876543213',
55 'sneha@gmail.com', 'Chennai'),
56
57 (105, 'Kiran Kumar', '9876543214',
58 'kiran@gmail.com', 'Hyderabad'),
59
60 (106, 'Anil Kumar', '9876543215',

```

Output

Action Output

#	Time	Action	Message
✓ 22	18:26:43	CREATE TABLE Account (Account_No INT PRIMARY...	0 row(s) affected
✓ 23	18:27:16	CREATE TABLE Bank_Transaction (Transaction_ID IN...	0 row(s) affected
✗ 24	18:27:55	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	Error Code: 1050. Table 'loan' e
✗ 25	18:29:03	INSERT INTO Customer (Customer_ID, Customer_Name, ...	Error Code: 1062. Duplicate ent

STEP 7: INSERT ACCOUNT DATA

MySQL Workbench

Bindhu Madhavi Local instanc...

File Edit View Query Database Server Tools Scripting Help

Navigator: eragate Functions(Week2)* Week1* sql command.txt* Trigeers (Week2) SQL File 15*

MANAGEMENT

- Server Status
- Client Connectic
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Par
- Administration

Information

No object selected

```

70 'lakshmi@gmail.com', 'Hyderabad'),
71
72 (110, 'Suresh Babu', '9876543219',
73 'suresh@gmail.com', 'Vijayawada');
74
75
76 INSERT INTO Account
77 (Account_No, Customer_ID, Account_Type, Balance, Branch)
78 VALUES
79 (10001, 101, 'Savings', 50000, 'Hyderabad'),
80 (10002, 102, 'Savings', 75000, 'Vijayawada'),
81 (10003, 103, 'Current', 120000, 'Bangalore'),
82 (10004, 104, 'Savings', 45000, 'Chennai'),
83 (10005, 105, 'Current', 90000, 'Hyderabad'),
84 (10006, 106, 'Savings', 65000, 'Delhi'),
85 (10007, 107, 'Current', 150000, 'Mumbai'),
86 (10008, 108, 'Savings', 35000, 'Pune'),
87 (10009, 109, 'Savings', 85000, 'Hyderabad'),
88 (10010, 110, 'Current', 110000, 'Vijayawada');
89
90
91

```

Output

Action Output

#	Time	Action	Message
23	18:27:16	CREATE TABLE Bank_Transaction (Transaction_ID IN...	0 row(s) affected
24	18:27:55	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	Error Code: 1050. Table 'loan' already
25	18:29:03	INSERT INTO Customer (Customer_ID, Customer_Name, ...	Error Code: 1062. Duplicate entry '101
26	18:30:21	INSERT INTO Account (Account_No, Customer_ID, Acco...	Error Code: 1452. Cannot add or upda

STEP 8: INSERT TRANSACTION DATA

MySQL Workbench

The screenshot shows the MySQL Workbench interface. The main editor displays a SQL query for inserting data into the Bank_Transaction table. The query is as follows:

```
87 (10009, 109, 'Savings', 85000, 'Hyderabad'),
88 (10010, 110, 'Current', 110000, 'Vijayawada');
89 INSERT INTO Bank_Transaction
90 (Account_No, Transaction_Type, Amount)
91 VALUES
92 (10001, 'DEPOSIT', 10000),
93 (10001, 'WITHDRAW', 5000),
94 (10002, 'DEPOSIT', 15000),
95 (10003, 'WITHDRAW', 20000),
96 (10004, 'DEPOSIT', 5000),
97 (10005, 'WITHDRAW', 10000),
98 (10006, 'DEPOSIT', 12000),
99 (10007, 'DEPOSIT', 25000),
100 (10008, 'WITHDRAW', 5000),
101 (10009, 'DEPOSIT', 20000),
102 (10010, 'WITHDRAW', 15000);
103
104
105
106
107
```

The left sidebar shows the 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System, Data Export, and Data Import/Restore. The 'INSTANCE' section shows Startup/Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section shows Dashboard, Performance Schema, and Administration. The bottom output window shows error messages for previous queries:

#	Time	Action	Message
24	18:27:55	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	Error Code: 1050. Table 'loan' already exists
25	18:29:03	INSERT INTO Customer (Customer_ID, Customer_Name, ...	Error Code: 1062. Duplicate entry
26	18:30:21	INSERT INTO Account (Account_No, Customer_ID, Acco...	Error Code: 1452. Cannot add or update data
27	18:31:35	INSERT INTO Bank_Transaction (Account_No, Transacti...	Error Code: 1452. Cannot add or update data

STEP 9: INSERT LOAN DATA

MySQL Workbench

Bindhu Madhavi Local instanc... x

File Edit View Query Database Server Tools Scripting Help

Navigator: Aggregate Functions(Week2)* Week1* sql command.txt* Trigeers (Week2) SQL File 15*

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Dashboard
- Administration

Information: No object selected

```

104 INSERT INTO Loan
105 (Loan_ID, Customer_ID, Loan_Type,
106 Loan_Amount, Interest_Rate)
107 VALUES
108 (501, 101, 'Home Loan', 5000000, 7.5),
109
110 (502, 102, 'Education Loan', 1000000, 6.5),
111
112 (503, 103, 'Car Loan', 800000, 8.2),
113
114 (504, 104, 'Personal Loan', 500000, 10.5),
115
116 (505, 105, 'Home Loan', 4000000, 7.2),
117
118 (506, 106, 'Car Loan', 900000, 8.5),
119
120 (507, 107, 'Business Loan', 3000000, 9.0),
121
122 (508, 109, 'Personal Loan', 600000, 10.0);
123
124
125

```

Output: Action Output

#	Time	Action	Message
25	18:29:03	INSERT INTO Customer (Customer_ID, Customer_Name, ...	Error Code: 1062. Duplicate entry '1
26	18:30:21	INSERT INTO Account (Account_No, Customer_ID, Acco...	Error Code: 1452. Cannot add or up
27	18:31:35	INSERT INTO Bank_Transaction (Account_No, Transacti...	Error Code: 1452. Cannot add or up
28	18:32:30	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, ...	Error Code: 1452. Cannot add or up

STEP 10: DISPLAY ORIGINAL TABLES

Bindhu Madhavi Local instanc... x

File Edit View Query Database Server Tools Scripting Help

Navigator: eregate Functions(Week2)* Week1* sql command.txt* Trigeers (Week2) SQL File 15*

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Per
- Administration

Information: No object selected

Limit to 1000 rows

```
118 (506, 106, 'Car Loan', 900000, 8.5),
119
120 (507, 107, 'Business Loan', 3000000, 9.0),
121
122 (508, 109, 'Personal Loan', 600000, 10.0);
123
124 • SELECT * FROM Customer;
```

Result Grid

	Customer_ID	Customer_Name	Phone	Email	City
▶	101	P.Bindhu Madhavi	9876543210	ravi@gmail.com	Hyderabad
	102	P.Bindhu Madhavi	9876543211	priya@gmail.com	Vijayawada
	103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
	104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
	105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

Customer 1 x Apply Revert

Output

Action Output

#	Time	Action	Message
✗	26 18:30:21	INSERT INTO Account (Account_No, Customer_ID, Acc...	Error Code: 1452. Cannot add or up
✗	27 18:31:35	INSERT INTO Bank_Transaction (Account_No, Transact...	Error Code: 1452. Cannot add or up
✗	28 18:32:30	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type,...	Error Code: 1452. Cannot add or up
✓	29 18:33:13	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned

MySQL Workbench

Bindhu Madhavi Local instanc... x

File Edit View Query Database Server Tools Scripting Help

Navigator: eregate Functions(Week2)* Week1* sql command.txt* Trigeers (Week2) SQL File 15

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Per
- Administration

Information

No object selected

Limit to 1000 rows

```
120      (507, 107, 'Business Loan', 3000000, 9.0),
121
122      (508, 109, 'Personal Loan', 600000, 10.0);
123
124 • SELECT * FROM Customer;
125 • SELECT * FROM Account;
126
```

Result Grid

	Account_No	Customer_ID	Account_Type	Balance	Branch
*	NULL	NULL	NULL	NULL	NULL

Account 2 x Apply

Output

Action Output

#	Time	Action	Message
✗ 27	18:31:35	INSERT INTO Bank_Transaction (Account_No, Transact...	Error Code: 1452. Cannot ad
✗ 28	18:32:30	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type,....	Error Code: 1452. Cannot ad
✓ 29	18:33:13	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
✓ 30	18:33:37	SELECT * FROM Account LIMIT 0, 1000	0 row(s) returned

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: Merge Functions(Week2)* Week1* sql command.txt* Triggers (Week2) SQL File 15*

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Dashboard
- Administration

Information

No object selected

Limit to 1000 rows

```
120      (507, 107, 'Business Loan', 3000000, 9.0),
121
122      (508, 109, 'Personal Loan', 600000, 10.0);
123
124 • SELECT * FROM Customer;
125 • SELECT * FROM Account;
126 • SELECT * FROM Bank_Transaction;
```

Result Grid

	Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
*	NULL	NULL	NULL	NULL	NULL

Bank_Transaction 3

Output

Action Output

#	Time	Action	Message
28	18:32:30	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type,...	Error Code: 1452. Cannot add or u
29	18:33:13	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
30	18:33:37	SELECT * FROM Account LIMIT 0, 1000	0 row(s) returned
31	18:34:06	SELECT * FROM Bank_Transaction LIMIT 0, 1000	0 row(s) returned

MySQL Workbench

Bindhu Madhavi Local instanc... x

File Edit View Query Database Server Tools Scripting Help

Navigator: eregate Functions(Week2)* Week1* sql command.txt* Trigeers (Week2) SQL File 15*

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutd
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Der
- Administration

Information

No object selected

```

121
122 (508, 109, 'Personal Loan', 600000, 10.0);
123
124 • SELECT * FROM Customer;
125 • SELECT * FROM Account;
126 • SELECT * FROM Bank_Transaction;
127 • SELECT * FROM Loan;

```

Result Grid

	Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
*	NULL	NULL	NULL	NULL	NULL

Filter Rows: Edit: Export/Imports: Result Grid Form Editor Field Types Query Stats

Loan 4 x Apply Revert Co

Output

Action Output

#	Time	Action	Message
✓ 29	18:33:13	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
✓ 30	18:33:37	SELECT * FROM Account LIMIT 0, 1000	0 row(s) returned
✓ 31	18:34:06	SELECT * FROM Bank_Transaction LIMIT 0, 1000	0 row(s) returned
✓ 32	18:34:36	SELECT * FROM Loan LIMIT 0, 1000	0 row(s) returned

PART A – SIMPLE VIEWS

STEP 11: CREATE A VIEW FOR ALL CUSTOMERS

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: eragate Functions(Week2)* Week1* sql command.txt* Triggers (Week2) SQL File 15*

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Dashboard
- Administration

Information: No object selected

```

127 • SELECT * FROM Loan;
128
129 • CREATE VIEW Customer_View AS
130 SELECT *
131 FROM Customer;
132
133 • SELECT * FROM Customer_View;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

	Customer_ID	Customer_Name	Phone	Email	City
▶	101	P.Bindhu Madhavi	9876543210	ravi@gmail.com	Hyderabad
	102	P.Bindhu Madhavi	9876543211	priya@gmail.com	Vijayawada
	103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
	104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
	105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad

Customer_View 5 x Read Only

Output

Action Output

#	Time	Action	Message
✓ 32	18:34:36	SELECT * FROM Loan LIMIT 0, 1000	0 row(s) returned
✓ 33	18:35:41	CREATE VIEW Customer_View AS SELECT * FROM Customer	0 row(s) affected
✗ 34	18:35:47	DISPLAY VIEW: SELECT * FROM Customer_View	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'SELECT * FROM Customer_View' at line 1
✓ 35	18:37:12	SELECT * FROM Customer_View LIMIT 0, 1000	5 row(s) returned

STEP 12: CREATE A VIEW FOR CUSTOMER BASIC INFORMATION

Bindhu Madhavi Local instanc... x

File Edit View Query Database Server Tools Scripting Help

Navigator: eregate Functions(Week2)* Week1* sql command.txt* Trigeers (Week2) SQL File 15*

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Per
- Administration

Information

No object selected

```

139 Customer_ID,
140 Customer_Name,
141 City
142 FROM Customer;
143 • SELECT * FROM Customer_Basic_View;
144
145

```

Limit to 1000 rows

Result Grid Filter Rows: Export: Wrap Cell Content:

	Customer_ID	Customer_Name	City
▶	101	P.Bindhu Madhavi	Hyderabad
	102	P.Bindhu Madhavi	Vijayawada
	103	Arjun Reddy	Bangalore
	104	Sneha Rao	Chennai
	105	Kiran Kumar	Hyderabad

Customer_Basic_View 6 x Read Only

Output

Action Output

#	Time	Action	Message
✓ 35	18:37:12	SELECT * FROM Customer_View LIMIT 0, 1000	5 row(s) returned
✗ 36	18:37:57	SELECT * FROM Customer_Basic_View LIMIT 0, 1000	Error Code: 1146. Table 'bank
✓ 37	18:38:15	CREATE VIEW Customer_Basic_View AS SELECT Cu...	0 row(s) affected
✓ 38	18:38:15	SELECT * FROM Customer_Basic_View LIMIT 0, 1000	5 row(s) returned

STEP 13: CREATE A VIEW FOR ACCOUNT INFORMATION

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: eregate Functions(Week2)* Week1* sql command.txt* Trigeers (Week2) SQL File 15*

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdo
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Per
- Administration

Information

No object selected

```

146 SELECT
147     Account_No,
148     Account_Type,
149     Balance,
150     Branch
151 FROM Account;
152 SELECT * FROM Account_View;
    
```

Result Grid

Account_No	Account_Type	Balance	Branch
------------	--------------	---------	--------

Account_View 7 x Read Only

Output

Action Output

#	Time	Action	Message
✓ 37	18:38:15	CREATE VIEW Customer_Basic_View AS SELECT Cu...	0 row(s) affected
✓ 38	18:38:15	SELECT * FROM Customer_Basic_View LIMIT 0, 1000	5 row(s) returned
✓ 39	18:39:47	CREATE VIEW Account_View AS SELECT Account_...	0 row(s) affected
✓ 40	18:39:47	SELECT * FROM Account_View LIMIT 0, 1000	0 row(s) returned

STEP 14: CREATE A VIEW FOR SAVINGS ACCOUNTS

MySQL Workbench

Bindhu Madhavi Local instanc... x

File Edit View Query Database Server Tools Scripting Help

Navigator: eregate Functions(Week2)* Week1* sql command.txt* Trigeers (Week2) SQL File 15*

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Per
- Administration

Information No object selected

```
153
154 • CREATE VIEW Savings_Account_View AS
155 SELECT *
156 FROM Account
157 WHERE Account_Type = 'Savings';
158 • SELECT * FROM Savings_Account_View;
159
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Account_No	Customer_ID	Account_Type	Balance	Branch
------------	-------------	--------------	---------	--------

Savings_Account_View 8 x Read Only

Output

Action Output

#	Time	Action	Message
✓ 39	18:39:47	CREATE VIEW Account_View AS SELECT Account_...	0 row(s) affected
✓ 40	18:39:47	SELECT * FROM Account_View LIMIT 0, 1000	0 row(s) returned
✓ 41	18:40:31	CREATE VIEW Savings_Account_View AS SELECT * F...	0 row(s) affected
✓ 42	18:40:31	SELECT * FROM Savings_Account_View LIMIT 0, 1000	0 row(s) returned

STEP 15: CREATE A VIEW FOR CURRENT ACCOUNTS

MySQL Workbench

Bindhu Madhavi Local instanc...

File Edit View Query Database Server Tools Scripting Help

Navigator: ...eregate Functions(Week2)* Week1* sql command.txt* Trigeers (Week2) SQL File 15*

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdo
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Dar
- Administration

Information :

No object selected

```

159
160 • CREATE VIEW Current_Account_View AS
161 SELECT *
162 FROM Account
163 WHERE Account_Type = 'Current';
164 • SELECT * FROM Current_Account_View;
165

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Account_No	Customer_ID	Account_Type	Balance	Branch
------------	-------------	--------------	---------	--------

Current_Account_View 9 x Read Only

Output :

Action Output

#	Time	Action	Message
✓ 41	18:40:31	CREATE VIEW Savings_Account_View AS SELECT * F...	0 row(s) affected
✓ 42	18:40:31	SELECT * FROM Savings_Account_View LIMIT 0, 1000	0 row(s) returned
✓ 43	18:41:20	CREATE VIEW Current_Account_View AS SELECT * FR...	0 row(s) affected
✓ 44	18:41:20	SELECT * FROM Current_Account_View LIMIT 0, 1000	0 row(s) returned

STEP 16: HIGH BALANCE ACCOUNT VIEW

MySQL Workbench

The screenshot shows the MySQL Workbench interface with the following components:

- Navigator:** Displays the database structure under 'Bindhu Madhavi Local instance...'. The 'MANAGEMENT' section is expanded, showing 'Server Status', 'Client Connections', 'Users and Privileges', 'Status and System', 'Data Export', and 'Data Import/Restore'. The 'INSTANCE' section shows 'Startup / Shutdown', 'Server Logs', and 'Options File'. The 'PERFORMANCE' section shows 'Dashboard', 'Performance Dashboard', and 'Administration'. The 'Information' section shows 'No object selected'.
- SQL Editor:** Contains the following SQL query:

```
167 SELECT
168     Account_No,
169     Customer_ID,
170     Account_Type,
171     Balance
172 FROM Account
173 WHERE Balance > 100000;
```
- Result Grid:** Displays the query results in a table with the following columns: Account_No, Customer_ID, Account_Type, and Balance.
- Output:** Shows the execution log with the following entries:

#	Time	Action	Message
43	18:41:20	CREATE VIEW Current_Account_View AS SELECT * FR...	0 row(s) affected
44	18:41:20	SELECT * FROM Current_Account_View LIMIT 0, 1000	0 row(s) returned
45	18:42:16	CREATE VIEW High_Balance_View AS SELECT Acco...	0 row(s) affected
46	18:42:16	SELECT * FROM High_Balance_View LIMIT 0, 1000	0 row(s) returned

STEP 17: HYDERABAD ACCOUNT VIEW

MySQL Workbench

The screenshot shows the MySQL Workbench interface with the following components:

- Left Sidebar:** Contains sections for **MANAGEMENT** (Server Status, Client Connections, Users and Privileges, Status and System, Data Export, Data Import/Restore), **INSTANCE** (Startup / Shutdown, Server Logs, Options File), and **PERFORMANCE** (Dashboard, Performance Schema, Administration).
- Top Panel:** Shows the instance name "Bindhu Madhavi Local instance..." and a menu bar with File, Edit, View, Query, Database, Server, Tools, Scripting, and Help.
- SQL Editor:** Displays the following SQL code:

```
176  
177 • CREATE VIEW Hyderabad_Account_View AS  
178 SELECT *  
179 FROM Account  
180 WHERE Branch = 'Hyderabad';  
181 • SELECT * FROM Hyderabad_Account_View;  
182
```
- Result Grid:** Shows the columns: Account_No, Customer_ID, Account_Type, Balance, Branch.
- Output Panel:** Displays the "Action Output" table with the following data:

#	Time	Action	Message
✓ 45	18:42:16	CREATE VIEW High_Balance_View AS SELECT Acco...	0 row(s) affected
✓ 46	18:42:16	SELECT * FROM High_Balance_View LIMIT 0, 1000	0 row(s) returned
✓ 47	18:43:05	CREATE VIEW Hyderabad_Account_View AS SELECT *...	0 row(s) affected
✓ 48	18:43:05	SELECT * FROM Hyderabad_Account_View LIMIT 0, 1000	0 row(s) returned

STEP 18: LOW BALANCE ACCOUNT VIEW

MySQL Workbench

Bindhu Madhavi Local instanc... x

File Edit View Query Database Server Tools Scripting Help

Navigator: eragate Functions(Week2)* Week1* sql command.txt* Trigeers (Week2) SQL File 15*

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Der
- Administration

Information No object selected

```

185 SELECT
186     Account_No,
187     Customer_ID,
188     Balance
189 FROM Account
190 WHERE Balance < 50000;
191 • SELECT * FROM Low_Balance_View;
  
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Account_No	Customer_ID	Balance
------------	-------------	---------

Low_Balance_View 12 x Read Only

Output :.....

Action Output

#	Time	Action	Message
✓ 47	18:43:05	CREATE VIEW Hyderabad_Account_View AS SELECT * ...	0 row(s) affected
✓ 48	18:43:05	SELECT * FROM Hyderabad_Account_View LIMIT 0, 1000	0 row(s) returned
✓ 49	18:44:02	CREATE VIEW Low_Balance_View AS SELECT Acco...	0 row(s) affected
✓ 50	18:44:02	SELECT * FROM Low_Balance_View LIMIT 0, 1000	0 row(s) returned

STEP 19: CUSTOMER + ACCOUNT VIEW

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: Aggregate Functions(Week2)* Week1* sql command.txt* Triggers (Week2) SQL File 15*

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report
- Administration

Information

No object selected

```

197 C.City,
198 A.Account_No,
199 A.Account_Type,
200 A.Balance,
201 A.Branch
202 FROM Customer C
203 JOIN Account A
  
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch
-------------	---------------	------	------------	--------------	---------	--------

Customer_Account_View 13 x Read Only

Output

Action Output

#	Time	Action	Message
49	18:44:02	CREATE VIEW Low_Balance_View AS SELECT Acco...	0 row(s) affected
50	18:44:02	SELECT * FROM Low_Balance_View LIMIT 0, 1000	0 row(s) returned
51	18:45:06	CREATE VIEW Customer_Account_View AS SELECT ...	0 row(s) affected
52	18:45:06	SELECT * FROM Customer_Account_View LIMIT 0, 1000	0 row(s) returned

STEP 20: HYDERABAD CUSTOMER ACCOUNT VIEW

MySQL Workbench

Bindhu Madhavi Local instance M... x Bindhu Madhavi Local instance M..x

File Edit View Query Database Server Tools Scripting Help

Navigator: Weeks2(TCL)* Joint and Operations* My Example* Praticse Example(Week2)* Agg

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard

Administration

Information

No object selected

```

215 FROM Customer C
216 JOIN Account A
217 ON C.Customer_ID = A.Customer_ID
218 WHERE A.Branch = 'Hyderabad';
219
220 • SELECT * FROM Hyderabad_Customer_Accounts;
221

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Customer_Name	City	Account_No	Account_Type	Balance
---------------	------	------------	--------------	---------

Hyderabad_Customer_Accounts... x Read Only

Output

Action Output

#	Time	Action	Message
51	18:45:06	CREATE VIEW Customer_Account_View AS SELECT ...	0 row(s) affected
52	18:45:06	SELECT * FROM Customer_Account_View LIMIT 0, 1000	0 row(s) returned
53	18:45:47	CREATE VIEW Hyderabad_Customer_Accounts AS SEL...	0 row(s) affected
54	18:46:09	SELECT * FROM Hyderabad_Customer_Accounts LIMIT ...	0 row(s) returned

STEP 21: CUSTOMER LOAN VIEW

MySQL Workbench

Bindhu Madhavi Local instance M... x Bindhu Madhavi Local instance M...x

File Edit View Query Database Server Tools Scripting Help

Navigator: Weeks2(TCL)* Joint and Operations* My Example* Praticse Example(Week2)* Agg SQL

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Administration

Information

No object selected

```

228      L.Loan_Amount,
229      L.Interest_Rate
230  FROM Customer C
231  JOIN Loan L
232  ON C.Customer_ID = L.Customer_ID;
233  SELECT * FROM Customer_Loan_View;
234

```

Limit to 1000 rows

Result Grid

Customer_ID	Customer_Name	City	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate
-------------	---------------	------	---------	-----------	-------------	---------------

Customer_Loan_View 15 x

Read Only

Output

Action Output

#	Time	Action	Message
53	18:45:47	CREATE VIEW Hyderabad_Customer_Accounts AS SEL...	0 row(s) affected
54	18:46:09	SELECT * FROM Hyderabad_Customer_Accounts LIMIT ...	0 row(s) returned
55	18:47:15	CREATE VIEW Customer_Loan_View AS SELECT C.C...	0 row(s) affected
56	18:47:32	SELECT * FROM Customer_Loan_View LIMIT 0, 1000	0 row(s) returned

Obiect Info

STEP 22: COMPLETE CUSTOMER BANKING VIEW

MySQL Workbench

Bindhu Madhavi Local instance... x Bindhu Madhavi Local instance M..x

File Edit View Query Database Server Tools Scripting Help

Navigator: Weeks2(TCL)* Joint and Operations* My Example* Praticse Example(Week2)* Agg

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Administration

Information

No object selected

248 ON C.Customer_ID = A.Customer_ID

249 LEFT JOIN Loan L

250 ON C.Customer_ID = L.Customer_ID;

251

252 • SELECT * FROM Customer_Banking_View;

253

254

Result Grid

	Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Brand
▶	101	P.Bindhu Madhavi	Hyderabad	NULL	NULL	NULL	NULL
	102	P.Bindhu Madhavi	Vijayawada	NULL	NULL	NULL	NULL
	103	Arjun Reddy	Bangalore	NULL	NULL	NULL	NULL
	104	Sneha Rao	Chennai	NULL	NULL	NULL	NULL
	105	Kiran Kumar	Hyderabad	NULL	NULL	NULL	NULL

Customer_Banking_View 16 x

Output

Action Output

#	Time	Action	Message
✓ 55	18:47:15	CREATE VIEW Customer_Loan_View AS SELECT C.C...	0 row(s) affected
✓ 56	18:47:32	SELECT * FROM Customer_Loan_View LIMIT 0, 1000	0 row(s) returned
✓ 57	18:48:32	CREATE VIEW Customer_Banking_View AS SELECT ...	0 row(s) affected
✓ 58	18:48:47	SELECT * FROM Customer_Banking_View LIMIT 0, 1000	5 row(s) returned

STEP 23: TOTAL BANK BALANCE

The screenshot shows the MySQL Workbench interface with two tabs: 'Bindhu Madhavi Local instance...' and 'Bindhu Madhavi Local instance M...'. The left sidebar contains a 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System, Data Export, and Data Import/Restore. Below this is an 'INSTANCE' section with Startup/Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section includes a Dashboard and Administration. The main area displays a SQL script with the following queries:

```

252 • SELECT * FROM Customer_Banking_View;
253 • CREATE VIEW Total_Bank_Balance AS
254   SELECT
255     SUM(Balance) AS Total_Balance
256   FROM Account;
257 • SELECT * FROM Total_Bank_Balance;
258

```

The 'Result Grid' shows a single row with the value 'NULL' under the column 'Total_Balance'. The 'Output' section at the bottom shows the 'Action Output' for the executed queries:

#	Time	Action	Message
✓ 57	18:48:32	CREATE VIEW Customer_Banking_View AS SELECT ...	0 row(s) affected
✓ 58	18:48:47	SELECT * FROM Customer_Banking_View LIMIT 0, 1000	5 row(s) returned
✓ 59	18:49:47	CREATE VIEW Total_Bank_Balance AS SELECT SU...	0 row(s) affected
✓ 60	18:50:02	SELECT * FROM Total_Bank_Balance LIMIT 0, 1000	1 row(s) returned

STEP 24: AVERAGE ACCOUNT BALANCE

MySQL Workbench

Bindhu Madhavi Local instance... x Bindhu Madhavi Local instance M..x

File Edit View Query Database Server Tools Scripting Help

Navigator: Weeks2(TCL)* Joint and Operations* My Example* Praticse Example(Week2)*

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdo
- Server Logs
- Options File

PERFORMANCE

- Dashboard

Administration

Information

No object selected

258 • CREATE VIEW Average_Account_Balance AS

259 SELECT

260 AVG(Balance) AS Average_Balance

261 FROM Account;

262

263 • SELECT * FROM Average_Account_Balance;

264

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Average_Balance
▶	NULL

Average_Account_Balance 18 x

Output

Action Output

#	Time	Action	Message
✓ 59	18:49:47	CREATE VIEW Total_Bank_Balance AS SELECT SU...	0 row(s) affected
✓ 60	18:50:02	SELECT * FROM Total_Bank_Balance LIMIT 0, 1000	1 row(s) returned
✓ 61	18:50:59	CREATE VIEW Average_Account_Balance AS SELECT ...	0 row(s) affected
✓ 62	18:51:03	SELECT * FROM Average_Account_Balance LIMIT 0, 10...	1 row(s) returned

Object Info

STEP 25: MAXIMUM ACCOUNT BALANCE

MySQL Workbench

Bindhu Madhavi Local instance... x Bindhu Madhavi Local instance M..x

File Edit View Query Database Server Tools Scripting Help

Navigator: Weeks2(TCL)* Joint and Operations* My Example* Praticse Example(Week2)* Agg

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard

Administration

Information

No object selected

```

264
265 • CREATE VIEW Maximum_Balance_View AS
266 SELECT
267     MAX(Balance) AS Maximum_Balance
268 FROM Account;
269 • SELECT * FROM Maximum_Balance_View;
270

```

Result Grid

Maximum_Balance
NULL

Maximum_Balance_View 19 x Read Only

Output

Action Output

#	Time	Action	Message
61	18:50:59	CREATE VIEW Average_Account_Balance AS SELECT ...	0 row(s) affected
62	18:51:03	SELECT * FROM Average_Account_Balance LIMIT 0, 10...	1 row(s) returned
63	18:52:01	CREATE VIEW Maximum_Balance_View AS SELECT ...	0 row(s) affected
64	18:52:06	SELECT * FROM Maximum_Balance_View LIMIT 0, 1000	1 row(s) returned

STEP 26: MINIMUM ACCOUNT BALANCE

MySQL Workbench

Bindhu Madhavi Local instance M... x Bindhu Madhavi Local instance M..x

File Edit View Query Database Server Tools Scripting Help

Navigator: Weeks2(TCL)* Joint and Operations* My Example* Praticse Example(Week2)* Agg

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdo
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Administration

Information

No object selected

```

271 • CREATE VIEW Minimum_Balance_View AS
272 SELECT
273     MIN(Balance) AS Minimum_Balance
274 FROM Account;
275 • SELECT * FROM Minimum_Balance_View;
276
277

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Minimum_Balance
NULL

Minimum_Balance_View 20 x Read Only

Output

Action Output

#	Time	Action	Message
63	18:52:01	CREATE VIEW Maximum_Balance_View AS SELECT ...	0 row(s) affected
64	18:52:06	SELECT * FROM Maximum_Balance_View LIMIT 0, 1000	1 row(s) returned
65	18:53:03	CREATE VIEW Minimum_Balance_View AS SELECT ...	0 row(s) affected
66	18:53:03	SELECT * FROM Minimum_Balance_View LIMIT 0, 1000	1 row(s) returned

Object Info

STEP 27: TOTAL NUMBER OF ACCOUNTS

MySQL Workbench

Bindhu Madhavi Local instance... x Bindhu Madhavi Local instance M...x

File Edit View Query Database Server Tools Scripting Help

Navigator: Weeks2(TCL)* Joint and Operations* My Example* Praticse Example(Week2)* Agg

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdo
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Administration
- Information

No object selected

```

276 • CREATE VIEW Account_Count_View AS
277 SELECT
278     COUNT(*) AS Total_Accounts
279 FROM Account;
280 • SELECT * FROM Account_Count_View;
281
282

```

Limit to 1000 rows

Result Grid

Total_Accounts
0

Export: Wrap Cell Content:

Account_Count_View 21 x Read On

Output: Action Output

#	Time	Action	Message
65	18:53:03	CREATE VIEW Minimum_Balance_View AS SELECT ...	0 row(s) affected
66	18:53:03	SELECT * FROM Minimum_Balance_View LIMIT 0, 1000	1 row(s) returned
67	18:54:08	CREATE VIEW Account_Count_View AS SELECT CO...	0 row(s) affected
68	18:54:08	SELECT * FROM Account_Count_View LIMIT 0, 1000	1 row(s) returned

STEP 28: BRANCH-WISE ACCOUNT COUNT

MySQL Workbench

Bindhu Madhavi Local instance... x Bindhu Madhavi Local instance M..x

File Edit View Query Database Server Tools Scripting Help

Navigator: Weeks2(TCL)* Joint and Operations* My Example* Praticse Example(Week2)* Agg

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdo
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Administration

Information

No object selected

300
301
302
303
304
305

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Branch	Number_of_Accounts

Branch_Account_Count 22 x

Output

Action Output

#	Time	Action	Message
67	18:54:08	CREATE VIEW Account_Count_View AS SELECT CO...	0 row(s) affected
68	18:54:08	SELECT * FROM Account_Count_View LIMIT 0, 1000	1 row(s) returned
69	18:55:03	CREATE VIEW Branch_Account_Count AS SELECT B...	0 row(s) affected
70	18:55:03	SELECT * FROM Branch_Account_Count LIMIT 0, 1000	0 row(s) returned

STEP 29: BRANCH-WISE TOTAL BALANCE

MySQL Workbench

Bindhu Madhavi Local instance M... x Bindhu Madhavi Local instance M..x

File Edit View Query Database Server Tools Scripting Help

Navigator: Weeks2(TCL)* Joint and Operations* My Example* Praticse Example(Week2)* Agg

MANAGEMENT

- Server Status
- Client Connection
- Users and Privileges
- Status and System
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard

Administration

Information

No object selected

```

286 GROUP BY Branch;
287 SELECT * FROM Branch_Account_Count;
288 CREATE VIEW Branch_Total_Balance AS
289 SELECT
290     Branch,
291     SUM(Balance) AS Total_Balance
292 FROM Account
  
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Branch	Total_Balance
--------	---------------

Branch_Total_Balance 23 x Read Only Cor

Output

Action Output

#	Time	Action	Message
69	18:55:03	CREATE VIEW Branch_Account_Count AS SELECT B...	0 row(s) affected
70	18:55:03	SELECT * FROM Branch_Account_Count LIMIT 0, 1000	0 row(s) returned
71	18:56:09	CREATE VIEW Branch_Total_Balance AS SELECT Br...	0 row(s) affected
72	18:56:09	SELECT * FROM Branch_Total_Balance LIMIT 0, 1000	0 row(s) returned

Object Info

STEP 30: ACCOUNT TYPE COUNT

MySQL Workbench

Bindhu Madhavi Local instance M... x Bindhu Madhavi Local instance M...x

File Edit View Query Database Server Tools Scripting Help

Navigator: Weeks2(TCL)* Joint and Operations* My Example* Praticse Example(Week2)* Agg

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdo
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Administration
- Information

No object selected

```

298 Account_Type,
299 COUNT(*) AS Number_of_Accounts
300 FROM Account
301 GROUP BY Account_Type;
302 SELECT * FROM Account_Type_Count;
303
304

```

Limit to 1000 rows

Result Grid

Account_Type	Number_of_Accounts
--------------	--------------------

Account_Type_Count 24 x

Output

Action Output

#	Time	Action	Message
71	18:56:09	CREATE VIEW Branch_Total_Balance AS SELECT Br...	0 row(s) affected
72	18:56:09	SELECT * FROM Branch_Total_Balance LIMIT 0, 1000	0 row(s) returned
73	18:58:41	CREATE VIEW Account_Type_Count AS SELECT Ac...	0 row(s) affected
74	18:58:41	SELECT * FROM Account_Type_Count LIMIT 0, 1000	0 row(s) returned

STEP 31: ACCOUNT TYPE TOTAL BALANCE

MySQL Workbench interface showing the creation of a view named `Account_Type_Balance`.

Query Editor:

```

303
304 CREATE VIEW Account_Type_Balance AS
305 SELECT
306     Account_Type,
307     SUM(Balance) AS Total_Balance
308 FROM Account
309 GROUP BY Account_Type;

```

Result Grid:

Account_Type	Total_Balance
--------------	---------------

Output:

#	Time	Action	Message
73	18:58:41	CREATE VIEW Account_Type_Count AS SELECT Ac...	0 row(s) affected
74	18:58:41	SELECT * FROM Account_Type_Count LIMIT 0, 1000	0 row(s) returned
75	18:59:41	CREATE VIEW Account_Type_Balance AS SELECT ...	0 row(s) affected
76	18:59:41	SELECT * FROM Account_Type_Balance LIMIT 0, 1000	0 row(s) returned

STEP 32: BRANCHES WITH MORE THAN ONE ACCOUNT

The screenshot shows the MySQL Workbench interface with the following components:

- Navigator:** Displays the database structure, including Server Status, Client Connections, Users and Privileges, Status and System, Data Export, and Data Import/Restore.
- INSTANCE:** Shows the instance status, including Startup / Shutdown, Server Logs, and Options File.
- PERFORMANCE:** Includes the Dashboard and Administration sections.
- Information:** A section indicating "No object selected".
- SQL Editor:** Contains the following SQL query:

```
315 Branch,  
316 COUNT(*) AS Number_of_Accounts  
317 FROM Account  
318 GROUP BY Branch  
319 HAVING COUNT(*) > 1;  
320 SELECT * FROM Multiple_Account_Branches;  
321
```
- Result Grid:** Displays the results of the query, showing two columns: Branch and Number_of_Accounts.
- Output:** Shows the execution log with the following entries:

#	Time	Action	Message
75	18:59:41	CREATE VIEW Account_Type_Balance AS SELECT ...	0 row(s) affected
76	18:59:41	SELECT * FROM Account_Type_Balance LIMIT 0, 1000	0 row(s) returned
77	19:08:04	CREATE VIEW Multiple_Account_Branches AS SELECT ...	0 row(s) affected
78	19:08:04	SELECT * FROM Multiple_Account_Branches LIMIT 0, 1...	0 row(s) returned

STEP 33: BRANCHES WITH TOTAL BALANCE ABOVE 100000

MySQL Workbench

Bindhu Madhavi Local instance... x Bindhu Madhavi Local instance M...x

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard

Administration

Information

No object selected

Weeks2(TCL)* Joint and Operations* My Example* Praticse Example(Week2)* Agg

Limit to 1000 rows

```

322 CREATE VIEW Rich_Branches AS
323 SELECT
324     Branch,
325     SUM(Balance) AS Total_Balance
326 FROM Account
327 GROUP BY Branch
328 HAVING SUM(Balance) > 100000;

```

Result Grid

Branch	Total_Balance
--------	---------------

Rich_Branches 27 x

Read Only Context

Output

Action Output

#	Time	Action	Message
77	19:08:04	CREATE VIEW Multiple_Account_Branches AS SELECT ...	0 row(s) affected
78	19:08:04	SELECT * FROM Multiple_Account_Branches LIMIT 0, 1...	0 row(s) returned
79	19:09:04	CREATE VIEW Rich_Branches AS SELECT Branch, ...	0 row(s) affected
80	19:09:04	SELECT * FROM Rich_Branches LIMIT 0, 1000	0 row(s) returned

STEP 34: DEPOSIT TRANSACTION VIEW

The screenshot displays the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System, Data Export, and Data Import/Restore. Below this is the 'INSTANCE' section with Startup/Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section includes a Dashboard and Administration link. The main editor window shows a SQL query in the 'Query' tab:

```
333 FROM Bank_Transaction
334 WHERE Transaction_Type = 'DEPOSIT';
335 SELECT * FROM Deposit_Transaction_View;
336
337
338
339
```

The query is executed, and the 'Result Grid' is displayed below the editor. It shows a table with the following columns: Transaction_ID, Account_No, Transaction_Type, Amount, and Transaction_Date. The table is currently empty.

The 'Output' tab at the bottom shows the 'Action Output' for the executed queries:

#	Time	Action	Message
79	19:09:04	CREATE VIEW Rich_Branches AS SELECT Branch, ...	0 row(s) affected
80	19:09:04	SELECT * FROM Rich_Branches LIMIT 0, 1000	0 row(s) returned
81	19:10:07	CREATE VIEW Deposit_Transaction_View AS SELECT *...	0 row(s) affected
82	19:10:07	SELECT * FROM Deposit_Transaction_View LIMIT 0, 1000	0 row(s) returned

STEP 35: WITHDRAWAL TRANSACTION VIEW

MySQL Workbench

Bindhu Madhavi Local instance M... x Bindhu Madhavi Local instance M... x

File Edit View Query Database Server Tools Scripting Help

Navigator: Weeks2(TCL)* Joint and Operations* My Example* Praticse Example(Week2)* Agg SQL

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard

Administration Information

No object selected

```

337 CREATE VIEW Withdrawal_Transaction_View AS
338 SELECT *
339 FROM Bank_Transaction
340 WHERE Transaction_Type = 'WITHDRAW';
341 SELECT * FROM Withdrawal_Transaction_View;
342
343

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
----------------	------------	------------------	--------	------------------

Withdrawal_Transaction_View 29 x Read Only Cont

Output

Action Output

#	Time	Action	Message
81	19:10:07	CREATE VIEW Deposit_Transaction_View AS SELECT *...	0 row(s) affected
82	19:10:07	SELECT * FROM Deposit_Transaction_View LIMIT 0, 1000	0 row(s) returned
83	19:10:57	CREATE VIEW Withdrawal_Transaction_View AS SELE...	0 row(s) affected
84	19:10:57	SELECT * FROM Withdrawal_Transaction_View LIMIT 0,...	0 row(s) returned

Object Info

STEP 36: TRANSACTION DETAILS WITH CUSTOMER

MySQL Workbench

Bindhu Madhavi Local instance... x Bindhu Madhavi Local instance M..x

File Edit View Query Database Server Tools Scripting Help

Navigator: Weeks2(TCL)* Joint and Operations* My Example* Praticse Example(Week2)* Agg SQL

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdo
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Administration

Information

No object selected

```

350 T.Transaction_Type,
351 T.Amount,
352 T.Transaction_Date
353 FROM Customer C
354 JOIN Account A
355 ON C.Customer_ID = A.Customer_ID
356 JOIN Bank_Transaction T

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Customer_Name	Account_No	Account_Type	Transaction_ID	Transaction_Type	Amount
---------------	------------	--------------	----------------	------------------	--------

Customer_Transaction_View 30 x Read Only

Output

Action Output

#	Time	Action	Message
83	19:10:57	CREATE VIEW Withdrawal_Transaction_View AS SELE...	0 row(s) affected
84	19:10:57	SELECT * FROM Withdrawal_Transaction_View LIMIT 0,...	0 row(s) returned
85	19:12:02	CREATE VIEW Customer_Transaction_View AS SELECT...	0 row(s) affected
86	19:12:02	SELECT * FROM Customer_Transaction_View LIMIT 0, 1...	0 row(s) returned

STEP 37: HIGH VALUE TRANSACTION VIEW

MySQL Workbench

Bindhu Madhavi Local instanc... x Bindhu Madhavi Local instance M..x

File Edit View Query Database Server Tools Scripting Help

Navigator: Weeks2(TCL)* Joint and Operations* My Example* Praticse Example(Week2)* Agg

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Administration

Information

No object selected

```

359
360 • CREATE VIEW High_Value_Transaction_View AS
361 SELECT *
362 FROM Bank_Transaction
363 WHERE Amount > 10000;
364 • SELECT * FROM High_Value_Transaction_View;
365

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
----------------	------------	------------------	--------	------------------

High_Value_Transaction_View 31 x

Output

Action Output

#	Time	Action	Message
85	19:12:02	CREATE VIEW Customer_Transaction_View AS SELECT...	0 row(s) affected
86	19:12:02	SELECT * FROM Customer_Transaction_View LIMIT 0, 1...	0 row(s) returned
87	19:12:51	CREATE VIEW High_Value_Transaction_View AS SELE...	0 row(s) affected
88	19:12:51	SELECT * FROM High_Value_Transaction_View LIMIT 0,...	0 row(s) returned

STEP 38: ACCOUNTS ORDERED BY BALANCE

The screenshot shows the MySQL Workbench interface with two tabs: 'Bindhu Madhavi Local instance...' and 'Bindhu Madhavi Local instance M...'. The left sidebar contains a 'Navigator' pane with sections for 'MANAGEMENT' (Server Status, Client Connections, Users and Privileges, Status and System, Data Export, Data Import/Restore), 'INSTANCE' (Startup / Shutdown, Server Logs, Options File), and 'PERFORMANCE' (Dashboard, Administration). The main editor area shows a SQL query in the 'Query' tab:

```

372 FROM Account
373 ORDER BY Balance DESC;
374 SELECT * FROM Balance_Ranking_View;
375
376
377
378

```

Below the query editor, the 'Result Grid' is visible, showing a table with the following columns: Account_No, Customer_ID, Account_Type, and Balance. The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message
87	19:12:51	CREATE VIEW High_Value_Transaction_View AS SELE...	0 row(s) affected
88	19:12:51	SELECT * FROM High_Value_Transaction_View LIMIT 0,...	0 row(s) returned
89	19:13:45	CREATE VIEW Balance_Ranking_View AS SELECT A...	0 row(s) affected
90	19:13:45	SELECT * FROM Balance_Ranking_View LIMIT 0 1000	0 row(s) returned

STEP 39: CUSTOMERS ORDERED BY NAME

MySQL Workbench

The screenshot shows the MySQL Workbench interface with the following components:

- Navigator:**
 - MANAGEMENT**
 - Server Status
 - Client Connectio
 - Users and Privile
 - Status and Syste
 - Data Export
 - Data Import/Res
 - INSTANCE**
 - Startup / Shutdo
 - Server Logs
 - Options File
 - PERFORMANCE**
 - Dashboard
 - Administration
 - Information
- SQL Editor:**

```

377 SELECT
378     Customer_ID,
379     Customer_Name,
380     City
381 FROM Customer
382 ORDER BY Customer_Name;
383 • SELECT * FROM Customer_Name_View;
        
```
- Result Grid:**

	Customer_ID	Customer_Name	City
▶	103	Arjun Reddy	Bangalore
	105	Kiran Kumar	Hyderabad
	101	P.Bindhu Madhavi	Hyderabad
	102	P.Bindhu Madhavi	Vijayawada
	104	Sneha Rao	Chennai
- Output:**

Customer_Name_View 33 x

Action Output

#	Time	Action	Message
✓ 89	19:13:45	CREATE VIEW Balance_Ranking_View AS SELECT A...	0 row(s) affected
✓ 90	19:13:45	SELECT * FROM Balance_Ranking_View LIMIT 0, 1000	0 row(s) returned
✓ 91	19:14:45	CREATE VIEW Customer_Name_View AS SELECT Cu...	0 row(s) affected
✓ 92	19:14:45	SELECT * FROM Customer_Name_View LIMIT 0, 1000	5 row(s) returned

STEP 40: LOAN INTEREST CALCULATION VIEW

The screenshot shows the MySQL Workbench interface with the following components:

- Navigator:**
 - MANAGEMENT:** Server Status, Client Connections, Users and Privileges, Status and System, Data Export, Data Import/Restore.
 - INSTANCE:** Startup / Shutdown, Server Logs, Options File.
 - PERFORMANCE:** Dashboard, Administration.
- Information:** No object selected.
- SQL Editor:**

```

389     Loan_Type,
390     Loan_Amount,
391     Interest_Rate,
392     (Loan_Amount * Interest_Rate / 100)
393     AS Annual_Interest
394 FROM Loan;
395 SELECT * FROM Loan_Interest_View;
            
```
- Result Grid:**

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate	Annual_Interest
---------	-------------	-----------	-------------	---------------	-----------------
- Output:**

#	Time	Action	Message
91	19:14:45	CREATE VIEW Customer_Name_View AS SELECT Cu...	0 row(s) affected
92	19:14:45	SELECT * FROM Customer_Name_View LIMIT 0, 1000	5 row(s) returned
93	19:15:46	CREATE VIEW Loan_Interest_View AS SELECT Loan...	0 row(s) affected
94	19:15:46	SELECT * FROM Loan_Interest_View LIMIT 0, 1000	0 row(s) returned

STEP 41: LOAN TOTAL AMOUNT VIEW

Bindhu Madhavi Local instance M... x Bindhu Madhavi Local instance M...x

File Edit View Query Database Server Tools Scripting Help

Navigator Weeks2(TCL)* Joint and Operations* My Example* Praticse Example(Week2)* Agg

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Administration

Information

No object selected

398 SELECT
399 Loan_ID,
400 Customer_ID,
401 Loan_Type,
402 Loan_Amount,
403 Interest_Rate,
404 Loan_Amount +

Limit to 1000 rows

Result Grid

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate	Total_Amount
---------	-------------	-----------	-------------	---------------	--------------

Filter Rows: Export: Wrap Cell Content:

Loan_Total_Amount_View 35 x Read On

Output

Action Output

#	Time	Action	Message
93	19:15:46	CREATE VIEW Loan_Interest_View AS SELECT Loan...	0 row(s) affected
94	19:15:46	SELECT * FROM Loan_Interest_View LIMIT 0, 1000	0 row(s) returned
95	19:17:11	CREATE VIEW Loan_Total_Amount_View AS SELECT ...	0 row(s) affected
96	19:17:11	SELECT * FROM Loan_Total_Amount_View LIMIT 0, 1000	0 row(s) returned

Obiect Info

